Lesson 3.4 - How to Calculate the Range, Midrange, and Coefficient of Range

> **Video:** How To Calculate the Range, MidRange, and Coefficient of Range by The Organic Chemistry Tutor
> **Chapter:** 3 - Central Tendency

Formulas

Let H = highest value and L = lowest value.

Quantity	Formula	Meaning
Range	$H - L$	Total spread of the data
Midrange	$\frac{H + L}{2}$	Halfway point between the extremes
Coefficient of range (CR)	$\frac{H - L}{H + L}$	Range relative to the size of the values (no units)

Example

In the given data set, the lowest value is $L = 5$ and the highest is $H = 45$.

(a) Range

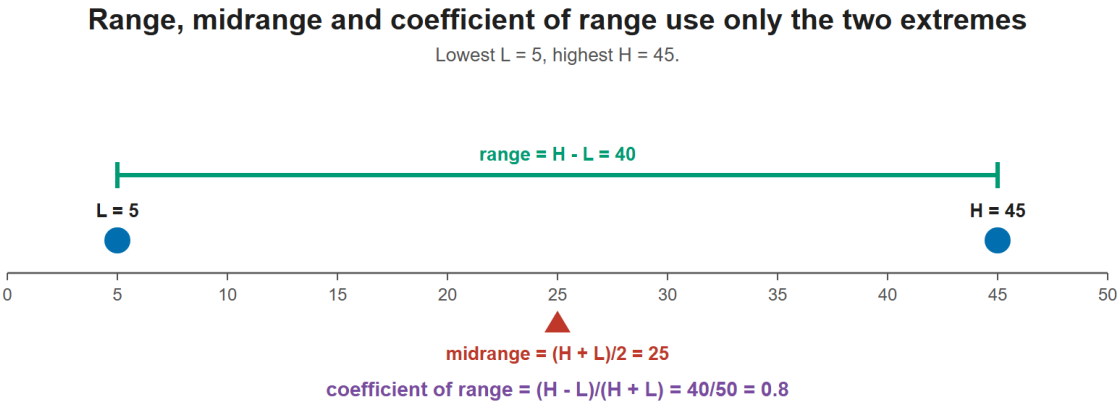
$$\text{range} = H - L = 45 - 5 = 40$$

(b) Midrange

$$\text{midrange} = \frac{H + L}{2} = \frac{45 + 5}{2} = \frac{50}{2} = 25$$

(c) Coefficient of range

$$\text{CR} = \frac{H - L}{H + L} = \frac{45 - 5}{45 + 5} = \frac{40}{50} = \frac{4}{5} = 0.8$$



All three measures come from just the two extreme values: the range is the distance between them, the midrange is the point halfway between them, and the coefficient of range compares the range with their sum.

Key takeaways

- All three use only the **highest** and **lowest** values, so they are sensitive to outliers.
- The midrange is a measure of center. The range and coefficient of range are measures of spread.
- The coefficient of range has no units, so it can compare the spread of data sets measured on different scales.